DAY 31 | PRE-READ

How Good Is Our AI, Really?

A light read before Day 31. About 15 minutes, no code required.

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Why read this?

Every customer will ask: "How do we know the AI is right?" Today you learn to answer with numbers instead of impressions, and to find the specific ways a system fails, including inventing facts, treating some users worse than others, and agents taking odd paths to an answer.

1. The tasting panel

A restaurant chain does not judge a new recipe by one chef tasting it once. A panel tastes many plates against a checklist: flavour, temperature, presentation. AI evaluation works the same way: a fixed set of realistic questions, a checklist of qualities, and scores you can compare over time.

2. Two places things go wrong

Stage	What can fail	Everyday version
Finding information	The right passage is never found	The librarian brings the wrong book
Writing the answer	The answer ignores or twists the passage	The student misquotes the right book

Measuring the two stages separately tells you what to fix.

3. Faithfulness

A faithful answer only says things the sources support. If an assistant adds "and employees get a free gym membership" when no document mentions it, that is a **hallucination**, even if it sounds plausible.

4. The route matters, not just the destination NEW (v4)

A taxi that reaches the airport by driving through three wrong towns still "arrived". For AI agents we check the whole **trajectory**: which tools it used, in what order, and whether it did anything risky on the way.

5. Right every time NEW (v4)

A support agent who is right 9 times out of 10 sounds good. But a customer who asks five times expects five right answers. **pass^k** measures how often an AI gets the same task right on every one of k tries.

6. Checking the checker NEW (v4)

Teams often use one AI model to grade another. These judges have habits, such as preferring longer answers, so we compare them with human grades on a sample first. This is called **calibration**.

Key idea: measure, then read the failures. Numbers tell you where to look; reading tells you why.

7. Think about these before class

- Write five questions a new employee would ask an HR assistant, with the correct answers.
- How would you know if an AI answer was made up?
- Which user groups might be served worse by an AI system in a telecom company?

Hakaru (測る): to measure